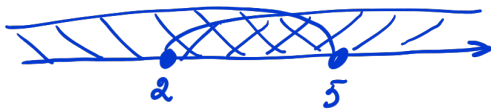


Sets

$$\begin{cases} x \geq 2 \\ x \leq 5 \end{cases}$$



$$\Rightarrow \text{solutions } 2 \leq x \leq 5 \\ \text{or } x \in [2, 5]$$

Naive definition of a Set

- A **set** is an unordered *collection* of **objects** called **elements** or *members* of the set.
- A set is said to contain its elements.
- We write $x \in S$ to denote that x is an element of the set S .

The set $\mathcal{V}$ of all vowels in the English alphabet can be written as

$$\mathcal{V} = \{a, e, i, o, u\}. \quad (\text{roster method})$$

The set of positive integers less than 100 can be denoted by $\{1, 2, \dots, 99, 100\}$

$\mathbb{N} = \{\underline{0}, 1, 2, 3, \dots\}$, the set of natural numbers

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers

$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, the set of positive integers

Set builder notation: $A = \{x \mid P(x)\}$, where P is some predicate.

↑
such that

Set of rational numbers: $\mathbb{Q} = \left\{ \frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{Z}, n \neq 0 \right\}$

$$[a, b] = \{x \mid a \leq x \leq b\}, [a, b) = \{x \mid a \leq x < b\}, \\ (a, b] = \{x \mid a < x \leq b\}, (a, b) = \{x \mid a < x < b\}$$

Definition

Two sets are equal if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{1, \cancel{X}, 3, 5\}$$

Sets $\{1, 3, 5\}$ and $\{5, 1, 3\}$ are equal. The order does not matter.

Sets can be weird, such as $\{1, 2, \text{blue}, \text{SOCIE}, 72, \text{apple}, \text{particles of the universe}\}$,

$\{a, b, c, \{a, b\}\}$, $\{A, B\}$, where A and B are sets too, empty set - denoted by $\emptyset$.

For example, $\{x \in \mathbb{R} \mid x^2 < 0\}$ is an empty set. The empty set can be thought of as an empty folder. The set consisting of just the empty set - $\{\emptyset\}$, can be thought of as a folder with exactly one folder inside, namely, the empty folder.

Why this set theory is “naïve”?

Note that the term object has been used in the definition of a set, based on the intuitive understanding of the German mathematician Georg Cantor in 1895. The theory that results from this definition, and the use of the intuitive notion that **for any property whatever, there is a set consisting of exactly the objects with this property**, leads to logical inconsistencies. This was shown by the English philosopher Bertrand Russell in 1902.

Let $P(A)$ = “set A is not an element of A ”.

For example, $P(\{1, 2\})$ is true since $\{1, 2\}$ is not an element of $\{1, 2\}$.

Consider this predicate P as a property, and by Cantor’s definition, collection of sets that satisfy property P must be a set. Let S be that set, that is

$$S = \{A \mid P(A)\} \quad (\text{meaning, } P(A) \text{ holds true}).$$

Then, what can we say about truth value of $P(S)$?

If $P(S) = 1$ then by construction S is a member of S , that contradicts $P(S) = 1$. ✗

If $P(S) = 0$, then S is an element of S . However, by construction of the set S we ✗ include only elements with $P(A) = 1$. Thus, $P(S) = 1$. Contradiction.

These logical inconsistencies can be avoided by building set theory beginning with axioms. In **von Neumann-Bernays-Gödel** axioms (NBG) **classes** are the basic objects in this theory, and a **set** is then defined to be a class that is an element of some other class.

Conclusion: Set of all sets is **not** a set. Collection of all sets is a class.

If interested have a look on the following theories on Wikipedia.

Zermelo-Fraenkel set theory, von Neumann-Bernays-Gödel set theory

Definition

The set A is a **subset** of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B
 $\Leftrightarrow \forall x(x \in A \rightarrow x \in B) \equiv \mathbf{1}$.

$A = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ and $\{S \mid S \text{ is a subset of } \{a, b\}\}$ are equal sets.

Theorem

For every set S we have (i) $\emptyset \subseteq S$ and (ii) $S \subseteq S$.

Proof: (i) need to prove $\forall x(x \in \emptyset \rightarrow x \in S) \equiv 1$, since $x \notin \emptyset \forall x$
(ii) $\forall x(x \in S \rightarrow x \in S) \equiv 1$

A subset A of S is called a **proper** subset of S if $A \neq S$.

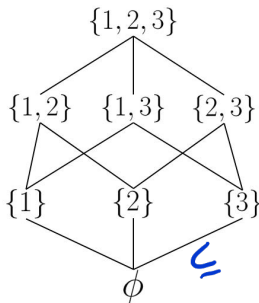
A is a proper subset of B if and only if $\underbrace{\forall x(x \in A \rightarrow x \in B)}_{\text{subset}} \wedge \exists x(x \in B \wedge x \notin A)$

Two sets A and B are equal if and only if $A \subseteq B$ and $B \subseteq A$.

$A = B$ if and only if $\forall x (x \in A \rightarrow x \in B) \wedge \forall x (x \in B \rightarrow x \in A)$

Find all subsets of a set $\{1, 2, 3\}$.

- There are 3 subsets with one elements: $\{1\}, \{2\}, \{3\}$
- There are 3 subsets with 2 elements: $\{1, 2\}, \{1, 3\}, \{2, 3\}$, ~~$\{2, 1\}$~~
- There are two more subsets: $\emptyset, \{1, 2, 3\}$



Answer Q1

Question: Find two sets A and B such that $A \in B$ and $A \subset B$.

$$\begin{array}{ccc} A = \{a\} & B = \{ \{a\}, a \} \\ & \uparrow \quad \uparrow \\ & A \in B \quad A \subset B \end{array}$$

The Cardinality of the Set

Let S be a set.

If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a **finite** set

n is the **cardinality** of S .

The cardinality of S is denoted by $|S|$.

A set is said to be **infinite** if it is not finite.

For example,

$$|\{a, u, e, i, o\}| = 5$$

$$|\{x | 0 \leq x \leq 8, x \text{ is odd}\}| = |\{1, 3, 5, 7\}| = 4$$

$$|\emptyset| = 0$$

$$|\{\{a\}, \{b, c, d\}\}| = 2$$

$$|\mathbb{N}| = \infty, \text{ since } \mathbb{N} \text{ is infinite}$$

$$A = \{a, u\}, B = \{a, u, e, i, o\}$$
$$A \subseteq B \quad |A| = 2 \quad |B| = 5$$

Answer Q2

finite

$\mathbb{N} \subset \mathbb{Z}$

If $A \subseteq B$, then $|A| \leq |B|$, but $|\mathbb{N}| = |\mathbb{Z}|$.

Power Set

Given a set S , the **power set** of S is the set of all subsets of the set S .
The power set of S is denoted by $\mathcal{P}(S)$.

In the previous lecture we saw that

$$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

$$\mathcal{P}(\emptyset) = \{\emptyset\}$$

$$\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$$

Question: Can you conclude that $A = B$ if A and B are two sets with the same power set? *Yes*

Cardinality of a power set

$$|\mathcal{P}(\{1, 2, 3\})| = 8 = 2^3$$

$$|\mathcal{P}(\emptyset)| = 1 = 2^0$$

$$|\mathcal{P}(\{\emptyset\})| = 2 = 2^1$$

Given a finite set S with $|S| = n$ we have $|\mathcal{P}(S)| = 2^n$

Answer Q3

Cartesian product

Ex: $\mathbb{R}^2$ - plane = $\mathbb{R} \times \mathbb{R}$

Given two sets A and B the **Cartesian product** of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$, $b \in B$.

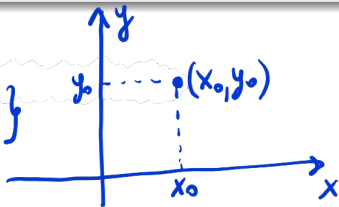
Let $A = \{a, b, c\}$ and $B = \{0, 1\}$. Then

$$A \times B = \{(a, 0), (a, 1), (b, 0), (b, 1), (c, 0), (c, 1)\}$$

The cardinality of $A \times B = 6 = 2 \cdot 3$

In general, $|A \times B| = |A| \cdot |B|$.

Note that $B \times A \neq A \times B$. Equality holds if $A = B$ or one of them is $\emptyset$.



The **Cartesian product** of the sets $A_1, A_2, \dots, A_n$ denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$, where a_i belongs to A_i for $i = 1, 2, \dots, n$. In other words,

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}.$$

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_n|.$$

Question from HW: Is it true that $A \times B \times C = (A \times B) \times C$? **no**

Homework:

- p. 124-125 of the book read the paragraph “Using Set Notation with Quantifiers” and “Truth Sets and Quantifiers”
- Solve *odd* exercises starting at p. 125 till #43.

$$\begin{cases} x \geq 2 \\ x \leq 5 \end{cases}$$

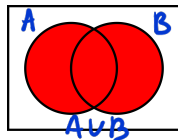


Set Operations

$$[2, 5] = (\infty, 5] \cap [2, \infty)$$

Union

- The **union** of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B (or in both).



- $A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$.
- Union of A and B is the **smallest** set that contains both A and B as subsets.

Examples: $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 3, 5, 2\}$

A = the set of all computer science majors

B = the set of all mathematics majors

$A \cup B$ = the set of students who are majoring either in mathematics or in computer science (or in both).

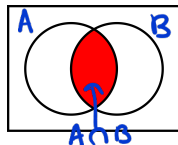
$P(x)$ = "Student x is computer science major"

$Q(x)$ = "Student x is mathematics major"

$P(x) \vee Q(x)$ is true for all students $x \in A \cup B$.

Intersection

- The **intersection** of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .



- $A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$.
- Intersection of two sets is the **largest** set which is a subset in both A and B .

Examples: $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$

A = the set of all computer science majors

B = the set of all mathematics majors

$A \cap B$ = are majoring in both mathematics and computer science.

$P(x)$ = "Student x is computer science major"

$Q(x)$ = "Student x is mathematics major"

$P(x) \wedge Q(x)$ is true for all students $x \in A \cap B$.

Answer Q4

Two sets are called **disjoint** if their intersection is the empty set.

$A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$ are disjoint, since $A \cap B = \emptyset$.

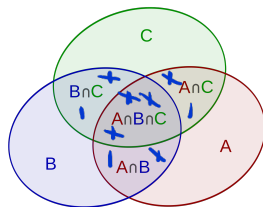
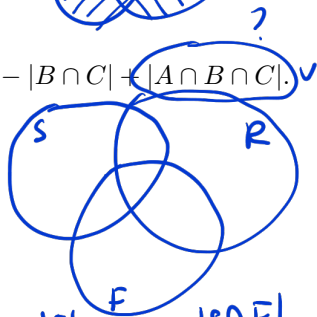
Inclusion-exclusion principle

For any two finite sets A and B

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets A, B, C , this is

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$



$$\begin{aligned} |S \cap R \cap F| &= \\ &= 2092 - 1232 - 879 - 114 \\ &\quad + 103 + 23 + 14 = \dots \end{aligned}$$

$|S|$ $|R|$ $|F \cap R|$

$|F|$ $|S \cap F|$

A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

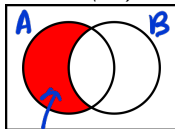
$|S \cap R \cap F|$

$|S \cap R|$

$|S \cap R \cap F|$

Difference

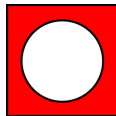
- The **difference** of A and B , denoted by $A - B$ (sometimes as $A \setminus B$), is the $(A \setminus B)$



$A \setminus B$

set containing those elements that are in A but not in B .

- $A - B = \{x \mid (x \in A) \wedge (x \notin B)\}$.
- The difference of A and B is called the **complement of B with respect to A** .
- Let U be the “universal” set.



The **complement** of the set A , denoted by $\overline{A}$, is $\overline{A} = U - A$.

Examples: $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$

What is $\{1, 3, 5\}$? If $U = \{1, 2, 3, \dots, 10\}$, then

$$\{1, 3, 5\} = \{2, 4, 6, 7, 8, 9, 10\}$$

If $U = \mathbb{N}$, then $\{1, 3, 5\} = \dots$ is different.

A = the set of all computer science majors

B = the set of all mathematics majors

$A - B$ = are majoring in computer science but not in mathematics.

$\overline{A}$ = are NOT majoring in computer science $\overline{U}$ - students of IUT

$P(x)$ = "Student x is computer science major"

$Q(x)$ = "Student x is mathematics major"

$P(x) \wedge \neg Q(x)$ is true for all students $x \in A - B$.

$\neg P(x)$ is true for all students $x \in \overline{A}$.

$$\begin{aligned} P \wedge \neg Q &= \neg(\neg P \vee Q) \\ &= \neg(\neg P \rightarrow Q) \end{aligned}$$

Exercise: $A - B = \underline{A} \cap \overline{B}$. Let $x \in \underline{A} - B \Leftrightarrow x \in A \wedge x \notin B \Rightarrow$

$\Leftrightarrow x \in A$ and $x \in \overline{B} \Leftrightarrow x \in A \cap \overline{B}$

Answer Q5

Familiar Laws

• $A \cap U \stackrel{=T}{=} A, A \cup \emptyset \stackrel{=F}{=} A$

Identity laws

• $A \cup U = U, A \cap \emptyset = \emptyset$

Domination laws

• $A \cup A = A, A \cap A = A$

Idempotent laws

• $\overline{(\overline{A})} = A$

Double negation law

• $A \cup B = B \cup A, A \cap B = B \cap A$

Commutative laws

• $(A \cup B) \cup C = A \cup (B \cup C)$

$(A \cap B) \cap C = A \cap (B \cap C)$

Associative laws

• $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Distributive laws

• $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$

$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$

De Morgan's laws

• $A \cup (A \cap B) = A$

$A \cap (A \cup B) = A$

Absorption laws

• $A \cup \overline{A} = U, A \cap \overline{A} = \emptyset$

Negation laws

Proof of De Morgan's Law: $x \in \overline{A \cap B} \equiv x \notin A \cap B \equiv \neg(x \in A \cap B) \equiv$

$\equiv \neg((x \in A) \wedge (x \in B)) \equiv \neg(x \in A) \vee \neg(x \in B) \equiv (x \notin A) \vee (x \notin B) \equiv x \in \overline{A} \cup \overline{B}.$

Hence, $\forall x(x \in \overline{A \cap B} \leftrightarrow x \in \overline{A} \cup \overline{B}) = \mathbf{T}$ and those sets are equal.

Another way to prove set identities

Let us prove distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$



When you pick an element $x \in U$, it might be related to sets A, B and C in 8 possible ways. Considering each, we build the following table.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

The 4th and 8th columns have identical values thus the sets under consideration are equal.

Answer Q6 - Q7

- The **union of a collection** of sets is the set that contains those elements that are members of at least one set in the collection.

- We use the notation $A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$.

- The **intersection of a collection** of sets is the set that contains those elements that are members of all the sets in the collection.

- We use the notation $A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$.

$$A_1 = \{1, 2, \dots, n\} \quad A_{10} = \{10, 11, \dots, n\} \quad A_n = \{n\}$$

$$A_2 = \{2, 3, \dots, n\} \quad n > 10$$



Let $A_i = \{i, i+1, i+2, \dots, n\}$. Then $\bigcap_{i=1}^{10} A_i = A_{10}$ and $\bigcap_{i=1}^n A_i = A_n = \{n\}$

Let $B_i = \{i, i+1, i+2, \dots\}$. Then $\bigcup_{i \geq 1} B_i = B_1 = \{1, 2, 3, \dots\}$

However, $\bigcap_{i \geq 1} B_i = \emptyset$

Let $C_i = \{1, 2, \dots, i\}$. Then $\bigcup_{i \geq 1} C_i = \mathbb{Z}^+$ and $\bigcap_{i \geq 1} C_i = C_1$

Answer Q7 8

Homework:

- p. 134 of the book read the paragraph “Computer representation of sets”
- Solve all *odd* exercises starting at p. 136.
- Solve *odd* exercises at p. 557, #1-#11.